

TECHNICAL REPORT · RENEWABLE ENERGY

Rooftop Solar PV System Design Basis & Cost Estimate

Marriott Hotel – Mezzanine Roof (Drawing D18102-DD-A-104)

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System capacity	185.4 kWp DC / 150 kW AC, grid-tied, no battery

Document Control

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Reference documents

Reference	Title
D18102-DD-A-104	Mezzanine floor plan incl. roof PV zones (architect)
D18102-DD-A-104 Layout1	Mezzanine plan with PV module layout, 10° tilt (this design)
PV_Layout_10deg_Mezzanine_Roof.dxf	PV layout CAD file – modules, tables, setbacks, dimensions

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1. Executive Summary

This report sets out the design basis, layout, energy yield and budget cost for a grid-tied rooftop solar photovoltaic (PV) system on the mezzanine-level roof of the Marriott Hotel, as shown on architectural drawing D18102-DD-A-104. The system has no battery storage and supplies the hotel's own load behind the utility meter.

185.4 kWp	274 MWh	USD 109k	3.4 yrs
DC capacity (299 × 620 Wp)	Year-1 generation	Budget EPC cost (0.59 USD/Wp)	Simple payback

Key findings

- **Capacity.** The two roof zones allocated for PV by the architect (≈1,150 m² gross, including the hatched +4.85 m slab strip) accommodate **299 modules of 620 Wp in 13 rows of 23**, giving **185.4 kWp** at 10° tilt with 800 mm maintenance aisles and 1.0 m edge setbacks.
- **Self-consumption.** The hotel consumes approximately **1,000,000 kWh per month** (average demand ≈1.37 MW). The PV peak output (≈150 kW AC) is around one-tenth of the daytime load, so **100 % of the solar energy is consumed on site**. No energy is exported and no battery is required.
- **Energy and savings.** Year-1 generation is about **274 MWh**, or 2.3 % of the annual consumption. At an avoided tariff of USD 0.12/kWh this saves about **USD 32,923 in year 1**.
- **Cost and return.** The budget turnkey (EPC) cost is **USD 108,636** excluding taxes, including 7 % contingency. Simple payback is **3.4 years**, project IRR 30 % and 25-year NPV (10 %) USD 208,322. The levelised cost of solar energy is about **USD 0.053/kWh**, below the grid tariff.
- **Path to 500 kWp.** The target of ~500 kWp cannot be met on this roof alone. An optional strip below the PV zone adds 42.8 kWp; the balance (≈270 kWp) requires further roof areas or car-park canopies (Section 8).

Recommendation

Proceed with the 185.4 kWp mezzanine roof system as Phase 1, subject to (i) a structural check of both slabs for an added dead load of about 20–25 kg/m² (including ballast) and wind uplift, and (ii) confirmation of the avoided tariff from recent utility bills. In parallel, survey other roof areas and the car park for Phase 2.

2. Site and Design Basis

2.1 Site and load

Facility	Marriott Hotel – twin towers with podium; PV on the mezzanine-level roof
Location	Uganda, equatorial (≈0–1° N); irradiation basis to be confirmed with site coordinates
Roof zones used	Upper block (+6.80 m SSL, 29.39 × 15.81 m) incl. hatched +4.85 m strip; lower block (architect's PV zone)
Annual electricity consumption	≈12,000,000 kWh (≈1,000,000 kWh/month)
Average / daytime demand	≈1,370 kW / ≈1,700 kW (hotel load profile assumed)
Grid connection	Behind-the-meter, LV main switchboard; zero-export control
Energy storage	None

2.2 Design criteria

Parameter	Value	Basis
Module	620 Wp bifacial TOPCon, 2382 × 1134 × 30 mm, ≈33 kg	Current Tier-1 mainstream size
Orientation / mounting	Portrait, 2 modules high (2P) per table	Matches roof depth
Tilt	10°	Low wind load, self-cleaning by rain
Table depth on plan	4,711 mm (2P); 2,346 mm (1P)	2 × 2382 + 20 mm, × cos 10°
Module clamp gap	20 mm	Mid-clamp
Aisle between tables	800 mm	No inter-row shading 09:00–15:00 all year (≥523 mm required); walkway for cleaning
Edge setback	1,000 mm (east side 1,742 mm)	Wind edge zone, safe access; clear of roof opening
Inverter DC voltage	1,100 V max	Standard C&I string inverter
Design standards	IEC 62548, IEC 60364-7-712, IEC 61215/61730, IEC 62109, IEC 62305	PV array, wiring, modules, inverters, lightning

3. PV Array Layout

The array occupies the two roof zones allocated for PV on drawing D18102-DD-A-104. Each row holds 23 modules (row length 26,522 mm). The upper block carries two 2P tables and one 1P table; the lower block carries four 2P tables.

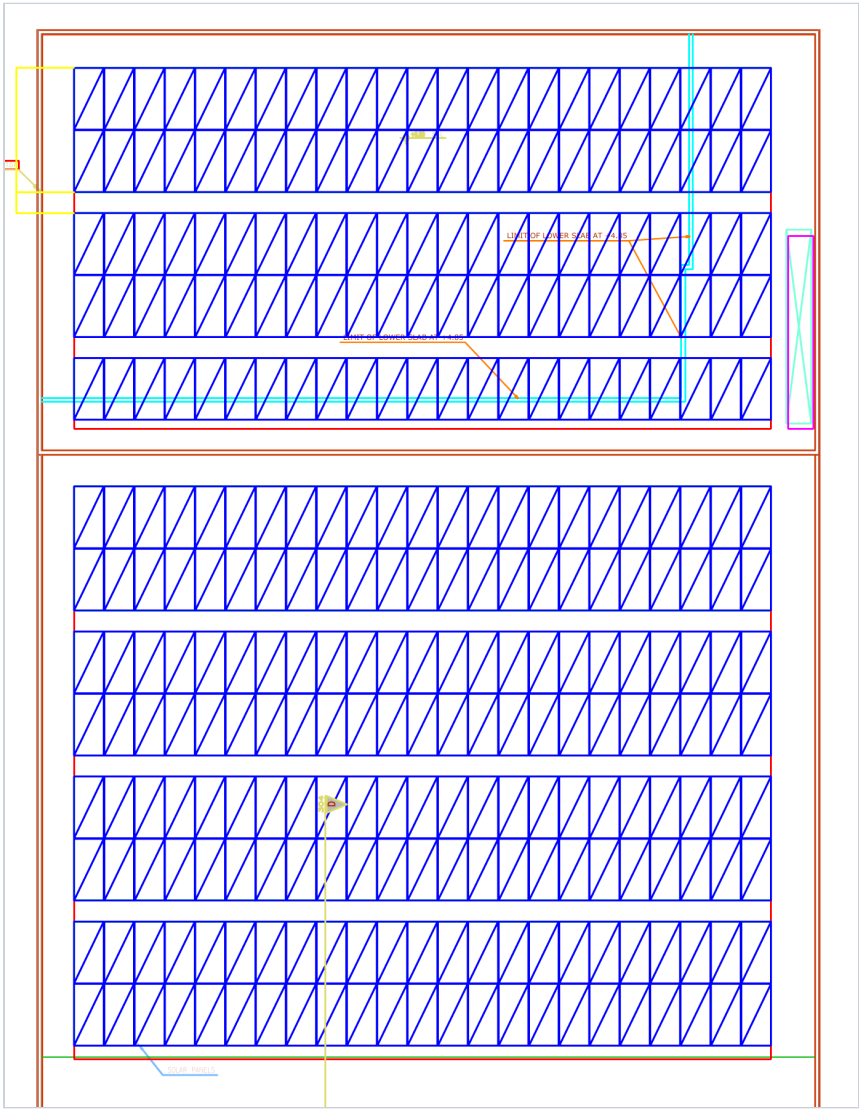


Figure 1 – PV module layout on the mezzanine roof (extract from D18102-DD-A-104 Layout1). Upper block at top, lower block below.

Zone	Tables	Rows	Modules	kWp
Upper block (+6.80 m, incl. hatched +4.85 m strip)	2 × 2P + 1 × 1P	5	115	71.3
Lower block (architect's PV zone)	4 × 2P	8	184	114.1
Total		**13**	**299**	**185.4**

Module area 807 m²; gross roof ≈1,150 m². Indicative added dead load ≈14 kg/m² for modules and structure, plus ballast to suit the wind design.

4. Electrical Design

4.1 Configuration

Item	Inverter INV-01	Inverter INV-02	Total
Rated AC power	75 kW	75 kW	150 kW
Strings	8 × 19 modules	3 × 19 + 5 × 18 modules	16 strings
Modules	152	147	299
DC capacity	94.2 kWp	91.1 kWp	185.4 kWp
DC/AC ratio	1.26	1.21	1.24

4.2 String voltage check

Check	Value	Limit	Result
Max. open-circuit voltage, 19 modules at 12 °C cell temperature (Voc ≈ 50.2 V)	954 V	≤ 1,100 V	OK
Min. MPP voltage, 18 modules at 65 °C cell temperature (Vmp ≈ 36.2 V)	652 V	≥ MPPT minimum (≈200 V)	OK
String current (Isc ≈ 16.1 A, ×1.25 for bifacial gain)	≈20 A	MPPT input rating	OK, max. 2 strings per MPPT

Module electrical values are typical for 620 Wp TOPCon modules. Re-check against the datasheet of the selected module and inverter.

4.3 Single-line arrangement

1. PV strings → 6 mm² double-insulated PV cable in UV-resistant trunking → inverters mounted on the roof in a shaded, ventilated location.
2. Inverters (400 V, 3-phase) → AC combiner panel with outgoing 250 A MCCB, Type II SPD and an isolator at the roof.
3. AC combiner panel → 4C × 120 mm² Cu XLPE/SWA (≈80 m, confirm route) → dedicated 250 A outgoing way on the main LV switchboard.
4. Zero-export controller with CTs on the incomer limits PV output so the site never exports, satisfying utility interface requirements.
5. Earthing: array frames bonded and connected to the building earth; integrated with the existing lightning protection system (IEC 62305).

5. Energy Yield

Specific yield (year 1, incl. losses)	1,480 kWh/kWp/yr
Performance ratio assumed	≈ 0.80 (temperature, soiling, cabling, inverter, availability)
Year-1 generation	274,362 kWh (≈ 22,864 kWh per month)
Degradation	1.0 % in year 1, then 0.4 %/yr
25-year generation	6,477 MWh
Self-consumption	100 % (PV peak ≈150 kW vs. daytime load ≈1,700 kW)
Share of hotel consumption	2.3 %

At the equator the sun crosses close to overhead and day length is nearly constant, so monthly output is fairly even. It is lowest in the main rainy seasons (March–May and October–November). Because the hotel load exceeds the PV output in every hour, cloudy days reduce savings but never cause curtailment.

The yield figure is a desktop estimate for an equatorial Ugandan site. A PVsyst simulation using the exact site coordinates and Meteonorm/PVGIS data is recommended at detailed design.

6. Cost Estimate (Bill of Quantities)

Budget turnkey (EPC) estimate in US dollars, based on 2026 East African market prices for commercial rooftop systems. Accuracy class ±15 %.

No.	Description	Qty	Rate	Amount (USD)
1	PV modules, 620 Wp bifacial TOPCon, Tier-1 (landed, cleared)	299 nos	USD 0.150/Wp	27,807
2	String inverters, 75 kW, 1,100 V, multi-MPPT (incl. comms)	2 nos	USD 4,500 each	9,000
3	Mounting structure, aluminium, 10° tilt, ballasted/anchored, incl. roof pads	185.4 kWp	USD 0.090/Wp	16,684
4	DC system: 6 mm² PV cable, MC4 connectors, DC isolators, SPD Type II	Lot	USD 0.040/Wp	7,415
5	AC system: AC combiner panel, 250 A MCCB, 4C×120 mm² Cu XLPE to main LV board (≈80 m), cable tray	Lot	Lump sum	9,500
6	Zero-export controller, CT/power meter, data logger & remote monitoring	1 set	Lump sum	3,500
7	Earthing, equipotential bonding & lightning protection integration	Lot	Lump sum	3,000
8	Roof walkways, safety lifeline & water point for cleaning	Lot	Lump sum	3,500
9	Installation, testing & commissioning (labour, supervision, lifting)	185.4 kWp	USD 0.060/Wp	11,123
10	Engineering: detailed design, structural check of slab, PVsyst, as-builts	Lot	Lump sum	4,000
11	Approvals & utility interface (ERA / UEDCL), O&M manuals, training	Lot	Lump sum	2,000
12	Transport, logistics & customs clearing	Lot	Lump sum	4,000
	Sub-total			**101,529**
	Contingency (7 %)			7,107
	Total budget EPC cost (excluding taxes)			**108,636**
	Specific cost			0.586 USD/Wp

Excludes VAT, withholding tax and import duties (solar equipment is generally zero-rated or exempt in Uganda; confirm with URA), roof waterproofing upgrades, structural strengthening if required, and any utility charges. Annual O&M (cleaning, inspection, monitoring) is estimated at USD 1,304/yr (≈1.2 % of capex), escalating at 3 %/yr.

7. Financial Analysis

Investment (EPC, excluding taxes)	USD 108,636
Avoided grid tariff (year 1) / escalation	USD 0.12/kWh / 2 % per year
Year-1 savings	USD 32,923
Inverter replacement (year 12)	USD 9,000
Simple payback	3.4 years
Project IRR (25 years, unlevered)	30.2 %
NPV at 10 % discount rate	USD 208,322
Levelised cost of energy (LCOE)	USD 0.053/kWh
25-year cumulative savings, net of O&M	USD 935,094

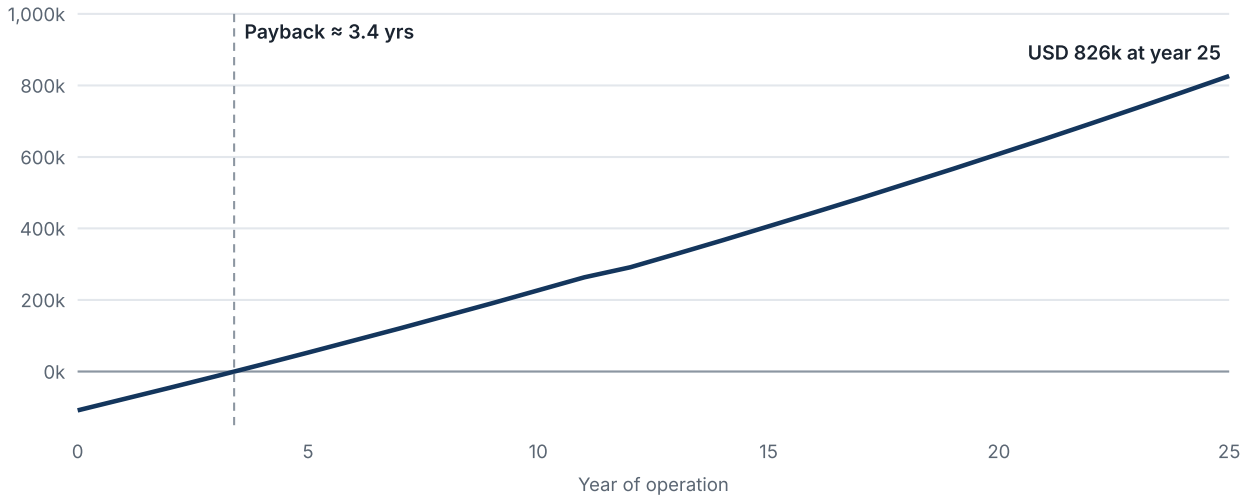


Figure 2 – Cumulative net cash flow (USD)

Tariff sensitivity

Avoided tariff (USD/kWh)	Payback (yrs)	IRR	NPV @10 % (USD)
0.09	4.6	23%	124,608
0.10	4.1	25%	152,513
0.12	**3.4**	**30%**	**208,322**
0.14	2.9	35%	264,131
0.16	2.5	40%	319,940

Annual cash flow (selected years, USD)

Year	Generation (MWh)	Tariff (USD/kWh)	Savings	O&M	Replacement	Net cash flow	Cumulative
0						-108,636	-108,636
1	274.4	0.120	32,923	1,304	–	31,619	-77,017
2	270.5	0.122	33,113	1,343	–	31,770	-45,246
3	269.4	0.125	33,640	1,383	–	32,257	-12,990
4	268.4	0.127	34,176	1,425	–	32,751	19,761
5	267.3	0.130	34,720	1,468	–	33,252	53,014
6	266.2	0.133	35,273	1,512	–	33,761	86,775
8	264.1	0.138	36,405	1,604	–	34,801	155,853
10	262.0	0.143	37,573	1,701	–	35,872	227,057
12	259.9	0.149	38,779	1,805	9,000	27,974	291,449
15	256.8	0.158	40,661	1,972	–	38,688	405,783
20	251.7	0.175	44,002	2,287	–	41,715	608,219
25	246.7	0.193	47,618	2,651	–	44,967	826,458

8. Expansion Options

The target of about 500 kWp exceeds what the mezzanine roof can hold. Because the hotel load is around ten times the PV output, every additional kWp would also be fully self-consumed. Expansion can be added in stages without a battery.

Option	Capacity	Year-1 energy	Budget cost (USD)	Remarks
Phase 1 – mezzanine roof (this report)	185.4 kWp	274 MWh	108,636	Ready for detailed design
Optional strip below PV zone (+4.85 m)	42.8 kWp	63 MWh	≈ 24,000	Confirm it is open roof and not needed for lift-core access
Other roofs (left podium roof ≈370 m², lobby roof)	≈ 65–120 kWp	≈ 95–180 MWh	≈ 38,000–68,000	Survey and structural check required
Car-park solar canopies	≈ 150–250 kWp	≈ 220–370 MWh	≈ 0.75–0.85 USD/Wp	Adds shaded parking; steel canopy cost
Indicative total	**≈ 450–600 kWp**	**≈ 650–890 MWh**		**≈ 5–7 % of hotel consumption**

9. Assumptions, Exclusions and Next Steps

9.1 Key assumptions

- Roof geometry is taken from drawing D18102-DD-A-104; the hatched +4.85 m slab strip is available for PV, as confirmed by the client.
- Both roof slabs can carry the PV dead load and wind loads without strengthening, subject to verification by the structural engineer.
- The hotel load is about 1,000,000 kWh per month with a continuous daytime demand well above 150 kW.
- The avoided tariff is USD 0.12/kWh escalating at 2 %/yr, to be confirmed from the latest utility bills (energy plus demand components).
- Prices are 2026 East African market levels in USD, excluding taxes. The exchange rate is not considered.

9.2 Exclusions

- Battery storage, diesel-generator synchronisation and export metering.
- Roof waterproofing renewal, structural strengthening, and civil works to the LV room.
- Utility network upgrades, permit fees and financing costs.

9.3 Recommended next steps

1. Obtain 12 months of utility bills or interval-meter data and confirm the tariff structure.
2. Commission a structural assessment of both slabs (dead load, ballast, wind uplift) and agree the roof fixing method with the architect.
3. Confirm the inverter location, AC cable route and spare way on the main LV switchboard.
4. Run a PVsyst simulation with site coordinates and the selected module and inverter datasheets.
5. Issue the tender package (layout, single-line diagram, specification, BOQ) to at least three qualified EPC contractors.
6. Start the Phase 2 survey of other roofs and the car park to move towards the 500 kWp target.